

**SAFETY DATA SHEET**

Creation Date 27-September-2014

Revision Date 26-March-2024

Revision Number 3

**1. Identification**

**Product Name** m-Toluoyl chloride

**Cat No. :** L02875

**CAS-No** 1711-06-4  
**Synonyms** 3-Methylbenzoyl chloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                |              |
|------------------------------------------------|--------------|
| <b>Flammable liquids</b>                       | Category 4   |
| <b>Skin Corrosion/Irritation</b>               | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>       | Category 1   |
| <b>Health Hazards Not Otherwise Classified</b> | Category 1   |
| <b>Lachrymator</b>                             |              |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Combustible liquid

Causes severe skin burns and eye damage

Lachrymator

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Wash contaminated clothing before reuse

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                   | CAS-No    | Weight % |
|-----------------------------|-----------|----------|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

**Inhalation**

Remove from exposure, lie down. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

**Ingestion**

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Clean mouth with water. Call a physician immediately.

**Most important symptoms/effects**

Difficulty in breathing. Causes burns by all exposure routes. . Symptoms of overexposure

**Notes to Physician**

may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation  
Treat symptomatically

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER OR FOAM                                                                                            |
| <b>Flash Point</b>                      | 76 °C / 168.8 °F                                                                                                    |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors. Phosgene. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
3

**Flammability**  
2

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                                                                                                   |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.                                                                                                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Storage.</b> | Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Store under an inert atmosphere. Incompatible Materials. Water. Strong bases. Alcohols.                                                                                      |

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles) Goggles

#### **Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                           |
|----------------------------------|---------------------------|
| Physical State                   | Liquid                    |
| Appearance                       | Light brown               |
| Odor                             | pungent                   |
| Odor Threshold                   | No information available  |
| pH                               | Not applicable            |
| Melting Point/Range              | No data available         |
| Boiling Point/Range              | 86 °C / 186.8 °F @ 5 mmHg |
| Flash Point                      | 76 °C / 168.8 °F          |
| Evaporation Rate                 | No information available  |
| Flammability (solid,gas)         | Not applicable            |
| Flammability or explosive limits |                           |
| Upper                            | No data available         |

|                                        |                                  |
|----------------------------------------|----------------------------------|
| Lower                                  | No data available                |
| Vapor Pressure                         | No information available         |
| Vapor Density                          | No information available         |
| Specific Gravity                       | 1.170                            |
| Solubility                             | Decomposes in contact with water |
| Partition coefficient; n-octanol/water | No data available                |
| Autoignition Temperature               | No information available         |
| Decomposition Temperature              | No information available         |
| Viscosity                              | No information available         |
| Molecular Formula                      | C8 H7 Cl O                       |
| Molecular Weight                       | 154.6                            |

## 10. Stability and reactivity

|                                  |                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                                                         |
| Stability                        | Moisture sensitive.                                                                                                                                                |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moisture.                                        |
| Incompatible Materials           | Water, Strong bases, Alcohols                                                                                                                                      |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Thermal decomposition can lead to release of irritating gases and vapors, Phosgene, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                           |
| Hazardous Reactions              | None under normal processing.                                                                                                                                      |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                   | LD50 Oral                 | LD50 Dermal | LC50 Inhalation |
|-----------------------------|---------------------------|-------------|-----------------|
| Benzoyl chloride, 3-methyl- | LD50 = 3440 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                   | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------|-----------|------------|------------|------------|------------|------------|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

|                                                   |                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                                                                                                      |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                  |

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

|                                      |                                                                 |
|--------------------------------------|-----------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available.         |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                       |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its volatility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3265                                    |
| <b>Proper Shipping Name</b> | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Technical Name</b>       | Benzoyl chloride, 3-methyl-               |
| <b>Hazard Class</b>         | 8                                         |
| <b>Packing Group</b>        | II                                        |

### TDG

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3265                                    |
| <b>Proper Shipping Name</b> | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                         |
| <b>Packing Group</b>        | II                                        |

### IATA

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3265                                    |
| <b>Proper Shipping Name</b> | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                         |
| <b>Packing Group</b>        | II                                        |

### IMDG/IMO

|                             |                                           |
|-----------------------------|-------------------------------------------|
| <b>UN-No</b>                | UN3265                                    |
| <b>Proper Shipping Name</b> | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                         |
| <b>Packing Group</b>        | II                                        |

## 15. Regulatory information

### International Inventories

| Component                   | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | -   | X    | X    | ACTIVE                                        | 216-976-8 | -      | -   |

| Component                   | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | X     | KE-23505 | X    | X    | X    | -    | -     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                   | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                   | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzoyl chloride, 3-methyl- | 1711-06-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Product Safety Department  
Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

**Creation Date**

27-September-2014

**Revision Date**

26-March-2024

**Print Date**

26-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**